

Mathematics for Engineers III

Y2 EPE

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Indicative content

Unit I : Fourier series and introduction to Fourier Transforms

- 1 Fourier series expansion
- 2 Fourier transform

Unit II: Partial Differential Equations including Boundary Value Problems

- 1 Formulation and Solution of standard types of first order equations
- 2 Lagrange's equation
- 3 Linear Homogeneous partial differential equations of second order with constant coefficients.
- 4 Classification of second order linear partial differential equations

Resources

- 1 Glyn James. (1999). Advanced modern engineering mathematics, 2nd edition Addison-Wesley.
- 2 K.A. Stroud. (1996). Further Engineering Mathematics, 3th edition Macmillan Press Ltd
- 3 Thomas H., Ronald J. (1977). Introductory Statistics Third edition, New York.
- 4 Robert V. Hogg. (2006). Probability and Statistical Inference. 7e edition. New Jersey.

Introduction

- Mathematicians of the eighteenth century, including **Daniel Bernoulli** and **Leonard Euler**, expressed the problem of the vibratory motion that had no solutions in terms of “elementary functions”.
- Their resolution of this difficult was to introduce infinite series of sine and cosine functions that satisfied the equations.
- In the early nineteenth century, **Joseph Fourier**, while studying the problem of heat flow, developed a cohesive theory of such series.
- One important advantage of Fourier series is that it can represent a function containing discontinuities, whereas Maclaurin's and Taylor's series require the function to be continuous throughout.

In this section we develop the **Fourier series expansion** of periodic functions, **Dirichlet's conditions**, **Half range sine and cosine series**

Periodic functions

- A function $f(x)$ is said to have a **period** or to be **periodic with period T** if for all x , $f(x) = f(x + T)$, where T is a positive constant.
- The least value of $T > 0$ is called the **least period** or simply **the period** of $f(x)$.
- Specifically, a function f is periodic with the period T if the graph of f is invariant under translation in the x -direction by the distance of T .
- A function that is not periodic is called **aperiodic**.

Example of Periodic function

Example

The function $f(x) = \sin(x)$ has periods $2\pi; 4\pi; 6\pi; \dots$; since $\sin(x + 2\pi); \sin(x + 4\pi); \sin(x + 6\pi); \dots$ all equal to $\sin(x)$. However, 2π is the least period or the period of the $\sin(x)$.

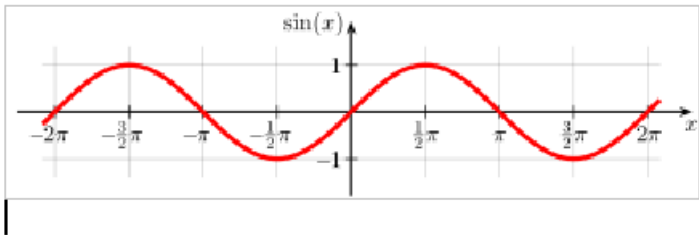


Figure: A graph of the sine function, showing two complete periods.

Examples

Example

The period of $\sin(nx)$ or $\cos(nx)$, where n is a positive integer, is $2\pi/n$.

Example

The period $\tan(x)$ is π .

Example

A constant has any positive number as period.

Example

Everyday examples are seen when the variable is **time**; for instance the hands of a clock or the phases of moon showing periodic behaviour. **Periodic motion** is motion in which the position(s) of the system are expressible as periodic functions, all with the *same* period.

Example

If we have the following periodic function $y = A \sin(nx)$, then A is an **amplitude**; **period** $= \frac{360^\circ}{n} = \frac{2\pi}{n}$, n cycle in 360° . The graphs of $y = A \cos(nx)$ have the same characteristics.

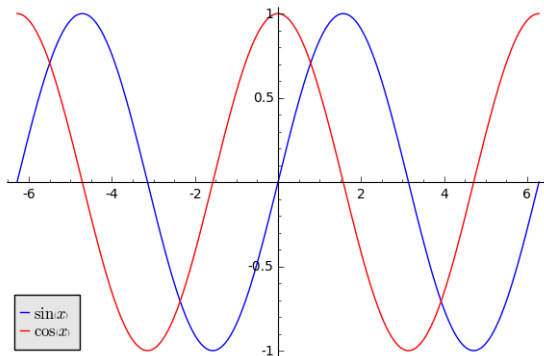


Figure: A graph of the sine and cosine functions, showing two complete periods.

Exercise

Example

Determine the amplitude and the period of the following periodic function

(a) $y = 3 \sin(5x)$

(b) $y = \sin(\frac{x}{2})$

(c) $Y = 6 \sin(\frac{2x}{3})$

Answers:

No.	Amplitude	Period
a)	3	$\frac{2\pi}{5}$
b)	1	4π
c)	6	3π

Harmonics

Harmonic functions

A function $f(x)$ is sometimes expressed as a series of a number of different sine components. The component with the largest period is the **first harmonics**, or **fundamental** of $f(x)$.

- $y = A_1 \sin(x)$ is the first harmonic or fundamental
- $y = A_2 \sin(2x)$ is the second harmonic
- $y = A_3 \sin(3x)$ is the third harmonic
- And in general $y = A_n \sin(nx)$ is the n^{th} harmonic, with amplitude A_n , and period $= \frac{360^\circ}{n} = \frac{2\pi}{n}$.

Non-sinusoidal periodic functions

A function can be periodic without being obviously sinusoidal in appearance.

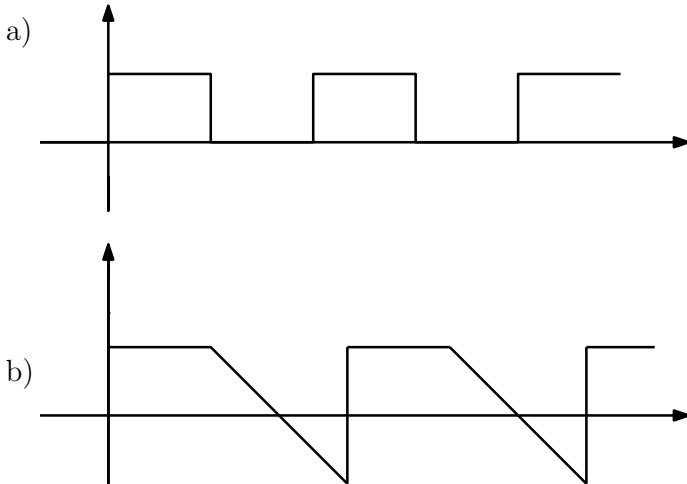


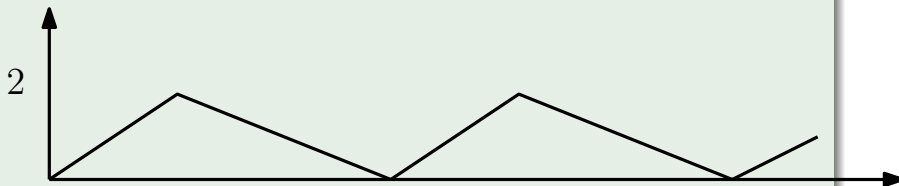
Figure: Example of periodic functions without being obviously sinusoidal

Analytical description periodic function.

Example

Sketch the graph of the following periodic function

$$f(x) = \begin{cases} x, & 0 < x < 2 \\ 3 - \frac{x}{2}, & 2 < x < 6 \end{cases}$$



$$f(x) = f(x + 6)$$

Figure: Analytical description periodic function

EXERCISES 1.1

1. Sketch the graphs of the following, inserting relevant values

$$a) f(x) = \begin{cases} 4, & 0 < x < 5 \\ 0, & 5 < x < 8 \\ f(x+8) \end{cases}$$

$$b) f(x) = \begin{cases} 3x - x^2, & 0 < x < 3 \\ f(x+3) \end{cases}$$

$$c) f(x) = \begin{cases} 2 \sin(x), & 0 < x < \pi \\ 0, & \pi < x < 2\pi \\ f(x+2\pi) \end{cases}$$

$$d) f(x) = \begin{cases} \frac{x^2}{4}, & 0 < x < 4 \\ 4, & 4 < x < 6 \\ 0, & 6 < x < 8 \\ f(x+8) \end{cases}$$

Fourier's theorem and the Fourier coefficients

Theorem

The theorem states that: *A periodic function that satisfies certain conditions which are Dirichlet conditions can be expressed as the sum of a number of sine functions of different amplitudes, phases and period.*

Definition

A Fourier series is an expansion of a periodic function $f(x)$ of period $T = 2\pi/k$ in which the base set is the set of sine functions, giving an expanded representation of the form

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \sin(nkx + \Phi_n) \quad (1)$$

where A_0 is a constant term.

Cont.....

Interpretations of the formula

$A_1, A_2, \dots, A_n, \dots$ denotes the **amplitudde** of the compound sine terms.
 $\Phi_1, \Phi_2, \dots, \Phi_n, \dots$ are constant auxiliary angles which are called **phase angle**,
 the terms $A_n \sin(nkx + \Phi_n)$ is called the **n th harmonic**.

Since,

$$\begin{aligned} A_n \sin(nkx + \Phi_n) &\equiv A_n \cos(\Phi_n) \sin(nkx) + A_n \sin(\Phi_n) \cos(nkx) \\ &\equiv a_n \cos(nkx) + b_n \sin(nkx) \end{aligned}$$

where $a_n = A_n \sin(\Phi)$, $b_n = A_n \cos(\Phi)$.

The expansion of a function in the form (1) had been used by **Bernouilli, D'Alembert and Euler** to solve problems associated with the vibration of strings. After Fourier postulated in 1807 that an arbitrary function could be represented by a trigonometric series.

Fourier series or Fourier Expansion

Let $f(x)$ be defined in the interval $(-L, L)$; and outside of this interval by $f(x + 2L) = f(x)$, i.e. $f(x)$ is $2L$ - periodic. It is through this avenue that a new function on an infinite set of real numbers is created from the image on $(-L, L)$. The Fourier series or Fourier expansion corresponding to $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right) \quad (2)$$

where the Fourier coefficients a_n and b_n are given as:

$$\begin{cases} a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \\ b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \end{cases} \quad \forall n = 0, 1, 2, \dots$$

Orthogonality Conditions for the sine and cosine functions

Notice that the Fourier coefficients are integrals. These are obtained starting with the series (2), and employing the following properties called orthogonality conditions:

Orthogonality conditions:

$$a) \int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 0, & \text{if } m \neq n \\ L, & \text{if } m = n \end{cases}$$

$$b) \int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 0, & \text{if } m \neq n \\ L, & \text{if } m = n \end{cases}$$

$$c) \int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = 0, \quad \text{where } m \text{ and } n \text{ can assume any positive integer values.}$$

Example

Example

Determine the Fourier coefficients a_0 and a_1 of the function given in equation (2) above.

Solution

a) To determine the Fourier coefficient a_0 , we integrate both sides of the Fourier series (2), i.e.,

$$\begin{aligned}\int_{-L}^L f(x) dx &= \int_{-L}^L \left[\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \right] dx \\&= \int_{-L}^L \frac{a_0}{2} dx + \int_{-L}^L \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) dx \\&= \int_{-L}^L \frac{a_0}{2} dx + \sum_{n=1}^{\infty} \left(a_n \int_{-L}^L \cos \frac{n\pi x}{L} dx + b_n \int_{-L}^L \sin \frac{n\pi x}{L} dx \right)\end{aligned}$$

Solution cont...

Finding the coefficient a_0

Since,

$$\int_{-L}^L \frac{a_0}{2} dx = a_0 L, \quad \int_{-L}^L \cos \frac{n\pi x}{L} dx = \int_{-L}^L \sin \frac{n\pi x}{L} dx = 0$$

. Thus,

$$\int_{-L}^L f(x) dx = a_0 L \quad \Rightarrow \quad a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

Finding the coefficient a_1

b) To determine Fourier coefficient a_1 , we multiply both sides of (2) by $\cos \frac{\pi x}{L}$ and then integrate.

$$\begin{aligned}\int_{-L}^L f(x) \cos \frac{\pi x}{L} dx &= \int_{-L}^L \cos \frac{\pi x}{L} \left[\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \right] dx \\ &= \frac{a_0}{2} \int_{-L}^L \cos \frac{\pi x}{L} dx + \sum_{n=1}^{\infty} \left(a_n \int_{-L}^L \cos \frac{n\pi x}{L} \cos \frac{\pi x}{L} dx + b_n \int_{-L}^L \sin \frac{n\pi x}{L} \cos \frac{\pi x}{L} dx \right)\end{aligned}$$

Thus,

$$\begin{aligned}\int_{-L}^L f(x) \cos \frac{\pi x}{L} dx &= a_1 \int_{-L}^L \cos \frac{\pi x}{L} \cos \frac{\pi x}{L} dx \\ &= a_1 L\end{aligned}$$

After using above orthogonality conditions. Therefore,

$$\int_{-L}^L f(x) \cos \frac{\pi x}{L} dx = a_1 L \quad \Rightarrow \quad a_1 = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{\pi x}{L} dx$$

Remark

Remark: If $L = \pi$, the series (2), and the coefficients a_n and b_n are particularly simple. The function in this case has the period 2π .

Example

Determine the Fourier series expansion of the following periodic functions of the period 2π :

a) $f(x) = x, \quad 0 < x < 2\pi$

b) $f(x) = x^2 + x, \quad -\pi < x < \pi$

c) $f(x) = \begin{cases} x, & 0 < x < \frac{\pi}{2} \\ \frac{\pi}{2}, & \frac{\pi}{2} < x < \pi \\ \pi - \frac{1}{2}x, & \pi < x < 2\pi. \end{cases}$

Solution for example a)

Fourier's coefficients determination:

$$a) a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} x dx = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{2\pi} = 2\pi$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx = \frac{1}{\pi} \int_0^{2\pi} x \cos nx dx$$

$$\Rightarrow a_n = \frac{1}{\pi} \left[x \frac{\sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{2\pi} = 0$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx = \frac{1}{\pi} \int_0^{2\pi} x \sin nx dx$$

$$\Rightarrow b_n = \frac{1}{\pi} \left[-\frac{x}{n} \cos nx + \frac{\sin nx}{n^2} \right]_0^{2\pi} = -\frac{2}{n}$$

Hence,

$$f(x) = \pi - \sum_{n=1}^{\infty} \frac{2}{n} \sin nx$$

Solution for example b)

$$\text{b) } f(x) = x^2 + x, \quad -\pi < x < \pi$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x^2 + x) dx = \frac{2}{3} \pi^2$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos \frac{n\pi x}{L} dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x^2 + x) \cos nx dx = \frac{1}{\pi} \frac{4\pi}{n^2} \cos \pi x = \frac{4}{n^2} (-1)^n$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin \frac{n\pi x}{L} dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x^2 + x) \sin nx dx = \frac{2}{n} \cos \pi x = \frac{2}{n} (-1)^n$$

Hence,

$$f(x) = \frac{2}{3} \pi^2 + \left[\sum_{n=1}^{\infty} \frac{2}{n} (-1)^n \left[\frac{2}{n} \cos nx + \sin nx \right] \right]$$

$$c) f(x) = \begin{cases} x, & 0 < x < \frac{1}{2}\pi \\ \frac{1}{2}\pi, & \frac{1}{2}\pi < x < \pi \\ \pi - \frac{1}{2}x, & \pi < x < 2\pi. \end{cases}$$

$$a_0 = \frac{1}{\pi} \int_{-L}^L f(x) dx = \frac{1}{\pi} \left[\int_0^{\frac{\pi}{2}} x dx + \int_{\frac{\pi}{2}}^{\pi} \frac{1}{2}\pi dx + \int_{\pi}^{2\pi} \left(\pi - \frac{1}{2}x\right) dx \right] = \frac{5}{8}\pi$$

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \\ &= \frac{1}{\pi} \left[\int_0^{\frac{\pi}{2}} x \cos nx dx + \int_{\frac{\pi}{2}}^{\pi} \frac{1}{2}\pi \cos nx dx + \int_{\pi}^{2\pi} \left(\pi - \frac{1}{2}x\right) \cos nx dx \right] \\ &= \frac{1}{\pi n^2} \left(2 \cos \frac{1}{2}n\pi - 3 + \cos n\pi \right) = \begin{cases} \frac{1}{2\pi n^2} [(-1)^n - 1] & n \text{ is even} \\ \frac{-1}{2\pi m^2} & n \text{ is odd} \end{cases} \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \\
 &= \frac{1}{\pi} \left[\int_0^{\frac{\pi}{2}} x \sin nx dx + \int_{\frac{\pi}{2}}^{\pi} \frac{1}{2} \pi \sin nx dx + \int_{\pi}^{2\pi} \left(\pi - \frac{1}{2}x \right) \sin nx dx \right] \\
 &= \frac{1}{\pi n^2} \sin \frac{1}{2} n\pi = \begin{cases} 0, & n \text{ is even} \\ \frac{(-1)^{(n-1)/2}}{\pi n^2} & n \text{ is odd} \end{cases}
 \end{aligned}$$

Hence,

$$\begin{aligned}
 f(x) &= \frac{5}{16} \pi - \frac{2}{\pi} \left(\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right) \\
 &\quad - \frac{2}{\pi} \left(\frac{\cos 2x}{2^2} + \frac{\cos 4x}{4^2} + \frac{\cos 6x}{6^2} + \dots \right) \\
 &\quad + \frac{1}{\pi} \left(\sin x + \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} + \dots \right)
 \end{aligned}$$

Remark: In the Fourier series corresponding to an odd function, only sine terms can be present. In the Fourier series corresponding to an even function, only cosine terms (and possibly a constant which we shall consider a cosine term) can be present.

Dirichlet's Conditions of Fourier's Series convergence

Suppose that

- ① $f(x)$ must be defined and single-valued on points in periodic interval $(-L, L)$;
- ② $f(x)$ must be continuous or have a finite number of finite discontinuities within a periodic interval $(-L, L)$ with period $2L$;
- ③ $f(x)$ and $f'(x)$ are piecewise continuous in periodic interval $(-L, L)$.

Then the series (??) with Fourier coefficients converges to

- a) $f(x)$ if x is a point of continuity;
- b) $\frac{f(x+0)+f(x-0)}{2}$ if x is a point of discontinuity.

Here $f(x+0)$ and $f(x-0)$ are the right and left-hand limits of $f(x)$ at x and represent

$$\lim_{\epsilon \rightarrow 0^+} f(x + \epsilon) \quad \text{and} \quad \lim_{\epsilon \rightarrow 0^-} f(x - \epsilon), \quad \text{respectively.}$$

Example

If the following functions are defined over the interval $-\pi < x < \pi$ and $f(x) = f(x + 2\pi)$, state whether or not each function can be represented by a Fourier series

- a) $f(x) = x^3$ b) $f(x) = 4x - 5$ c) $f(x) = \frac{2}{x}$ d) $f(x) = \frac{1}{x-5}$
 e) $f(x) = \tan x$ f) $f(x) = y$ where $x^2 + y^2 = 9$.

Answers: A given function can be represented by a Fourier series sufficiently satisfying the above three conditions,

No.	Statement
a)	Yes
b)	Yes
c)	No: Because is infinitely discontinuity at $x = 0$.
d)	Yes
e)	No: Because is infinitely discontinuity at $x = \frac{\pi}{2}$
f)	No: Because has two values

Half Range Fourier Cosine and Sine series

A half range Fourier sine or cosine series is a series in which only sine terms or only cosine terms are present, respectively. When a half range series corresponding to a given function is desired, the function is generally, defined in the interval $(0, L)$ which is half of the interval $(-L, L)$ thus accounting for the name **half range** and then the function is specified as **odd** or **even**, so that it is clearly defined in the other half of the interval namely, $(-L, 0)$.

For a half range cosine series

For a function $f(x)$ defined only over the finite interval $0 \leq x \leq L$ its **even periodic extension** $F(x)$ is the even periodic function defined by

$$F(x) = \begin{cases} f(x); & 0 < x < L \\ f(-x); & -L < x < 0 \end{cases} \quad F(x + 2L) = F(x)$$

If $f(x)$ satisfies Dirichlet's conditions in the interval $0 < x < L$, then since it is an even function of period $2L$, then even periodic extension $F(x)$ will have a convergent Fourier series representation consisting of cosine terms only and given by

$$F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}, \quad \text{where} \quad (3)$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx, \quad n = 0, 1, 2, 3, \dots$$

For a Half Range Sine Series

For a function $f(x)$ defined only over the finite interval $0 < x < L$, its **odd periodic extension** $G(x)$ is the odd periodic function defined by

$$G(x) = \begin{cases} f(x); & 0 < x < L \\ -f(-x); & -L < x < 0 \end{cases} \quad G(x + 2L) = G(x)$$

If $f(x)$ satisfies Dirichlet's conditions in the interval $0 < x < L$, then since it is an odd function of period $2L$, then odd periodic extension **$G(x)$** will have a convergent Fourier series representation consisting of sine terms only and given by

$$G(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \quad \text{where} \quad (4)$$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L}, \quad n = 1, 2, 3, \dots$$

Example

Consider the following function defined in the interval $0 < x < 4$, $f(x) = x$, obtain:

- a half range cosine series expansion.
- a half range sine series expansion

Solutions:

a) Half-range cosine series expansion. Define the periodic function

$$F(x) = \begin{cases} f(x) = x; & 0 < x < 4 \\ f(-x) = -x; & -4 < x < 0 \end{cases} \quad F(x+8) = F(x)$$

Then, since $F(x)$ is an even periodic function with period 8, it has a convergent Fourier series expansion given by (3). Taking $L = 4$, we have the Fourier's Coefficients determination as :

$$2 \int_0^4$$

$$1 \int_0^4$$

$$\begin{aligned}a_n &= \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx = \frac{2}{4} \int_0^4 x \cos \frac{n\pi x}{4} dx \\&= \frac{1}{2} \left[\frac{4x \sin n\pi x}{n\pi} + \frac{16 \cos n\pi x}{(n\pi)^2} \right]_0^4 \\&= \frac{8}{(n\pi)^2} (\cos n\pi - 1) = \begin{cases} 0, & \text{for } n \text{ is even} \\ -\frac{16}{(n\pi)^2}, & \text{for } n \text{ is odd} \end{cases}\end{aligned}$$

Hence,

$$F(x) = 2 - \frac{16}{\pi^2} \left(\cos \frac{1}{4}\pi x + \frac{1}{3^2} \cos \frac{3}{4}\pi x + \frac{1}{5^2} \cos \frac{5}{4}\pi x + \dots \right) \quad \text{or}$$

$$F(x) = 2 - \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{1}{4}(2n-1)\pi x$$

Since $F(x) = f(x)$ for $0 < x < 4$; it follows that this Fourier series is representative of $f(x)$ within this interval.

Thus the **half-range cosine series expansion** of $f(x)$ is

$$F(x) = x = 2 - \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{1}{4}(2n-1)\pi x, \quad \text{for } 0 < x < 4.$$

b) Half-range sine series expansion

. Define the periodic function

$$G(x) = \begin{cases} f(x) = x; & 0 < x < 4, \\ -f(x) = -x; & -4 < x < 0, \end{cases} \quad G(x+8) = G(x).$$

Then, since $G(x)$ is an odd periodic function with period 8, it has a convergent Fourier series expansion given by (4), we have the Fourier's coefficients determination as:

$$\begin{aligned}b_n &= \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx = \frac{2}{4} \int_0^4 x \sin \frac{n\pi x}{4} dx \\&= \frac{1}{2} \left[\frac{4x}{n\pi} \cos \frac{1}{4} n\pi x + \frac{16}{(n\pi)^2} \sin \frac{1}{4} n\pi x \right]_0^4 = -\frac{8}{\pi} \cos n\pi = -\frac{8}{n\pi} (-1)^n\end{aligned}$$

Hence,

$$\begin{aligned}G(x) &= \frac{8}{\pi} \left(\sin \frac{1}{4} \pi x - \frac{1}{2} \sin \frac{1}{2} \pi x + \frac{1}{3} \sin \frac{3}{4} \pi x + \dots \right) \quad \text{or} \\G(x) &= \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{1}{4} n\pi x\end{aligned}$$

Since $G(x) = f(x)$ for $0 < x < 4$, it follows that this Fourier series is representative of $f(x)$ within this interval. Thus the **half-range sine series expansion of $f(x)$** is

$$f(x) = x = \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{1}{4} n\pi x, \quad \text{for } 0 < x < 4.$$

Parseval's Identity

If a_n and b_n are the Fourier coefficients corresponding to $f(x)$ and if $f(x)$ satisfies the Dirichlet conditions, then

$$\frac{1}{L} \int_{-L}^L (f(x))^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Example

Consider the following function defined in the interval $0 < x < 4$, $f(x) = x$, verify the Parseval's identity.

$$\frac{1}{2} \int_0^4 x^2 dx = \frac{4^2}{2} + \sum_{n=1}^{\infty} \left(\left(\frac{-8}{(n\pi)^2} \right)^2 + 0^2 \right), \quad \text{for } n \text{ is an odd}$$

Example

$$\frac{1}{2} \int_0^4 x^2 dx = 8 + \frac{64}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^4}, \quad \text{for } n \text{ is odd}$$

$$\frac{32}{3} = 8 + \frac{64}{\pi^4} \left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots \right)$$

This implies that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^4}{24}$$

The Fourier transform provides a way of expanding functions on the whole real line $\mathbb{R} = (-\infty, \infty)$ as (continuous) superpositions of the basic oscillatory functions $e^{i\omega x}$, $\omega \in \mathbb{R}$ in much the same way that Fourier series are

Definition

A function f is said to be **absolutely integrable** over an interval (a, b) if and only if we can legitimately write

$$\int_a^b |f(x)| < \infty$$

The Fourier Integral

Let us assume the following condition on $f(x)$:

- 1 $f(x)$ satisfies the Dirichlet conditions in every finite interval $(-L, L)$;
- 2 $\int_{-\infty}^{\infty} |f(x)| dx$ converges, i.e., $f(x)$ is absolutely integrable in $(-\infty, \infty)$;

Then Fourier's integral theorem states that the **Fourier integral of a function f** is

$$\varphi(x) = \int_0^{\infty} \{A(\alpha) \cos \alpha x + B(\alpha) \sin \alpha x\} d\alpha \quad \text{where} \quad (5)$$

$$\begin{cases} A(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \alpha x dx \\ B(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \alpha x dx \end{cases}$$

$A(\alpha)$ and $B(\alpha)$ with $-\infty < \alpha < \infty$ are generalization of the Fourier coefficients a_n and b_n . The right-hand side of (5) is also called a **Fourier integral expansion of $f(x)$**

Equivalent Forms of Fourier's Integral Theorem

Fourier's integral theorem can also be written in the forms

$$\varphi(x) = \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{u=-\infty}^{\infty} f(u) \cos \alpha(x-u) du d\alpha$$

We can write $\cos \alpha(x-u) = e^{i(\alpha x - \alpha u)}$, then the above expression becomes:

$$\begin{aligned} \varphi(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\alpha x} d\alpha \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} du \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) e^{i\alpha(x-u)} d\alpha du. \end{aligned} \quad (6)$$

These results can be simplified somewhat if $f(x)$ is either an odd or an even function, and we have:

$$\varphi(x) = \frac{2}{\pi} \int_0^{\infty} \left(\int_0^{\infty} f(u) \cos \alpha x \cos \alpha u du \right) d\alpha, \quad \text{if } f(x) \text{ is an even} \quad (7)$$

$$\varphi(x) = \frac{2}{\pi} \int_0^{\infty} \left(\int_0^{\infty} f(u) \sin \alpha x \sin \alpha u du \right) d\alpha, \quad \text{if } f(x) \text{ is an odd} \quad (8)$$

Equation (7) and (8) formulas are called the **Fourier cosine integral**, and the **Fourier sine integral**, respectively.

An entity of importance in evaluating integrals and solving differential and integral equations, then we can put in (7) in the following form:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\alpha x} \underbrace{\left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} du \right\}}_{F(i\alpha)} d\alpha$$

The Fourier transform

From (6) it follows that

$$F(i\alpha) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\alpha x} dx \quad \text{and} \quad (9)$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(i\alpha) e^{i\alpha x} d\alpha. \quad (10)$$

Where the function $F(\alpha)$ is called the **Fourier transform** of $f(x)$ and is sometimes written $F(i\alpha) = F\{f(x)\}$.

The function $f(x)$ is the **inverse Fourier transform** of $F(x)$ and is written $f(x) = F^{-1}\{f(x)\}$.

Example

Determine the Fourier transform of f if

$$f(x) = \begin{cases} e^{-x}, & \text{for } x \geq 0 \\ e^{2x}, & \text{for } x < 0 \end{cases}$$

$$\begin{aligned} F(i\alpha) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\alpha x} dx \\ &= \frac{1}{\sqrt{2\pi}} \left\{ \int_{-\infty}^0 e^{2x} e^{-i\alpha x} dx + \int_0^{\infty} e^{-x} e^{-i\alpha x} dx \right\} \\ &= \frac{1}{\sqrt{2\pi}} \left\{ \int_{-\infty}^0 e^{2x-i\alpha x} dx + \int_0^{\infty} e^{-(x+i\alpha x)} dx \right\} \\ &= \frac{1}{\sqrt{2\pi}} \left\{ \frac{e^{(2-i\alpha)x}}{2-i\alpha} \Big|_{x \rightarrow -\infty}^{x \rightarrow 0^-} - \frac{e^{-(1+i\alpha)x}}{1+i\alpha} \Big|_{x \rightarrow 0^+}^{x \rightarrow \infty} \right\} \\ &= \frac{1}{\sqrt{2\pi}} \left\{ \frac{1}{2-i\alpha} + \frac{1}{1+i\alpha} \right\} = \frac{1}{\sqrt{2\pi}} \left\{ \frac{\alpha^2 - 2 + i\alpha(5 + 2\alpha^2)}{(4 + \alpha^2)(1 + \alpha^2)} \right\} \end{aligned}$$

Hence,

$$F(i\alpha) = \frac{1}{\sqrt{2\pi}} \left\{ \frac{\alpha^2 - 2 + i\alpha(5 + 2\alpha^2)}{(4 + a\alpha^2)(1 + \alpha^2)} \right\}$$

Remarks:

- If $f(x)$ is an even function, equation (8) yields

$$\begin{cases} F_c(i\alpha) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos \alpha x \, dx \\ f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty F_c(i\alpha) \cos \alpha x \, d\alpha \end{cases}$$

and we call $F_c(i\alpha)$ and $f(x)$ **Fourier cosine transforms** of each other.

- If $f(x)$ is an odd function, equation (8) yields

$$\begin{cases} F_s(i\alpha) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin \alpha x \, dx \\ f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty F_s(i\alpha) \sin \alpha x \, d\alpha \end{cases}$$

and we call $F_s(i\alpha)$ and $f(x)$ **Fourier sine transforms** of each other.

When the product of Fourier transforms is considered, a new concept called **convolution** comes into being, and in conjunction with it, a new pair (function and its Fourier transform) arises. In particular, if $F(i\alpha)$ and $G(i\alpha)$ are the Fourier transforms of f and g , respectively, and the convolution of f and g is defined to be

$$f * g = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(u)g(x-u)du \quad (11)$$

Then

$$F(i\alpha)G(i\alpha) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-i\alpha u} f * g du \quad (12)$$

$$f * g = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{i\alpha x} F(i\alpha)G(i\alpha) d\alpha \quad (13)$$

where in both (13) and (13) the convolution $f * g$ is a function of x . Now equate the representations of $f * g$ expressed in (13) and (14), i.e.,

$$\begin{aligned} f * g &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-i\alpha x} f(u) g(x - u) du \\ &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{i\alpha x} F(i\alpha) G(i\alpha) d\alpha \end{aligned}$$

and let the parameter x be zero, then

$$\int_{-\infty}^{\infty} f(u) g(-u) du = \int_{-\infty}^{\infty} F(i\alpha) G(i\alpha) d\alpha \quad (14)$$

Properties of Fourier Transforms for simple functions

The linearity property

Linearity property is a fundamental property of the Fourier transform, and may be stated as follows:

If $f(x)$ and $g(x)$ are functions having Fourier transforms $F(i\alpha)$ and $G(i\alpha)$, respectively, and if β and γ are constants, then

$$\mathcal{F}\{\beta f(x) + \gamma g(x)\} = \beta \mathcal{F}\{f(x)\} + \gamma \mathcal{F}\{g(x)\} = \beta F(i\alpha) + \gamma G(i\alpha)$$

As a consequence of this, we say that the Fourier transform operator $\mathcal{F}$ is a **linear operator**.

Proof of linearity property

Proof.

By definition we have:

$$\begin{aligned}\mathcal{F}\{\beta f(x) + \gamma g(x)\} &= \int_{-\infty}^{\infty} [\beta f(x) + \gamma g(x)] e^{-i\alpha x} dx \\ &= \beta \int_{-\infty}^{\infty} f(x) e^{-i\alpha x} dx + \gamma \int_{-\infty}^{\infty} g(x) e^{-i\alpha x} dx \\ &= \beta F(i\alpha) + \gamma G(i\alpha)\end{aligned}$$

Clearly the linearity property also applies to the inverse transform operator $\mathcal{F}^{-1}$ □

If the function $f(x)$ has a Fourier transform $F(i\alpha)$, then by (10)

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(i\alpha) e^{i\alpha x} d\alpha$$

Differentiating with respect to x gives

$$\frac{df}{dx} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{d}{dx} [F(i\alpha) e^{i\alpha x}] d\alpha = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (i\alpha) F(i\alpha) e^{i\alpha x} d\alpha$$

Imply that $\frac{df}{dx}$ is the inverse Fourier transform of $(i\alpha)F(i\alpha)$.

In other words

$$\mathcal{F} \left\{ \frac{df}{dx} \right\} = (i\alpha)F(i\alpha)$$

Repeating the argument n times, it follows that

$$\mathcal{F} \left\{ \frac{d^n f}{dx^n} \right\} = (i\alpha)^n F(i\alpha) \quad (15)$$

Example

Show that if the time signals $y(t)$ and $u(t)$ have the Fourier transforms $Y(i\alpha)$ and $U(i\alpha)$ respectively, and if

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 7y(t) = 3 \frac{du(t)}{dt} + 7u(t).$$

Then $Y(i\alpha) = G(i\alpha)U(i\alpha)$ for some function $G(i\alpha)$?

Solution: Taking Fourier transform of above differential equation, then we have:

$$\mathcal{F} \left\{ \frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 7y(t) \right\} = \mathcal{F} \left\{ 3 \frac{du(t)}{dt} + 7u(t) \right\}$$

Cont...

which, on using the linearity property, reduces to

$$\mathcal{F}\left\{\frac{d^2y(t)}{dt^2}\right\} + 3\mathcal{F}\left\{\frac{dy(t)}{dt}\right\} + 7\mathcal{F}\{y(t)\} = 3\mathcal{F}\left\{\frac{du(t)}{dt}\right\} + 7\mathcal{F}\{u(t)\}$$

Then, from (15) we get:

$$(i\alpha)^2 Y(i\alpha) + 3(i\alpha)Y(i\alpha) + 7Y(i\alpha) = 3(i\alpha)U(i\alpha) + 7U(i\alpha)$$

That is,

$$(7 - \alpha^2 + 3i\alpha)Y(i\alpha) = (7 + 3i\alpha)U(i\alpha)$$

Giving

$$Y(i\alpha) = G(i\alpha)U(i\alpha)$$

, where

$$G(i\alpha) = \frac{7 + 3i\alpha}{7 - \alpha^2 + 3i\alpha}$$

Time-Shift Property

If a function $f(t)$ has Fourier transform $F(i\alpha)$, then what is the Fourier transform of the shifted version of $f(t)$, defined by $g(t) = f(t - \tau)$?

Solution:

$$\mathcal{F}\{g(t)\} = \int_{-\infty}^{\infty} g(t)e^{-i\alpha t} dt = \int_{-\infty}^{\infty} f(t - \tau)e^{-i\alpha t} dt$$

Making the substitution $x = t - \tau$, we have

$$\begin{aligned}\mathcal{F}\{g(t)\} &= \int_{-\infty}^{\infty} f(x)e^{-i\alpha(x+\tau)} dx \\ &= e^{-i\alpha\tau} \int_{-\infty}^{\infty} f(x)e^{-i\alpha x} dx = e^{-i\alpha\tau} F(i\alpha).\end{aligned}$$

That is,

$$\mathcal{F}\{f(t - \tau)\} = e^{-i\alpha\tau} F(i\alpha) \quad (16)$$

Cont...

The result (16) is known as the **time-shift property**, and implies that delaying a signal by a time τ causes its Fourier transform to be multiplied by $e^{-i\alpha\tau}$.

Since

$$|e^{-i\alpha\tau}| = |\cos \alpha\tau - i \sin \alpha\tau| = 1$$

we have

$$|e^{-i\alpha\tau} F(i\alpha)| = |F(i\alpha)|$$

Indicating that the amplitude spectrum of $f(t - \tau)$ is identical with that of $f(t)$.

Frequency-Shift Property

Suppose that a function $f(t)$ has Fourier transform $F(i\alpha)$. Then, from the definition of Fourier transform we calculate the Fourier transform of $g(t) = e^{i\alpha_0 t} f(t)$ as

$$\begin{aligned}\mathcal{F}\{g(t)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\alpha_0 t} f(t) e^{-i\alpha t} dt = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i(\alpha - \alpha_0)t} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\tilde{\alpha}t} dt, \quad \text{where } \tilde{\alpha} = \alpha - \alpha_0 \\ &= F(i\tilde{\alpha}).\end{aligned}$$

Thus,

$$\mathcal{F}\{e^{i\alpha_0 t} f(t)\} = F[i(\alpha - \alpha_0)] \quad (17)$$

Example

Example

Determine the frequency spectrum of signal $g(t) = f(t) \cos \alpha_c t$

Solution: As $\cos \alpha_c t = \frac{1}{2}[e^{i\alpha_c t} + e^{-i\alpha_c t}]$

$$\begin{aligned}\mathcal{F}\{g(t)\} &= \mathcal{F}\left\{\frac{1}{2}f(t)[e^{i\alpha_c t} + e^{-i\alpha_c t}]\right\} \\ &= \frac{1}{2} \left[\mathcal{F}\{e^{i\alpha_c t} f(t)\} + \mathcal{F}\{e^{-i\alpha_c t} f(t)\} \right]\end{aligned}$$

Use the property (17), then we get:

$$\frac{1}{2}F[i(\alpha - \alpha_c)] + \frac{1}{2}F[i(\alpha + \alpha_c)]$$

The Symmetry Property

We are establish the exact form of symmentry as:

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(i\alpha) e^{-i\alpha t} d\alpha$$

Or, equivalently, by changing the dummy variable in the integration

$$\begin{aligned}\sqrt{2\pi}f(t) &= \int_{-\infty}^{\infty} F(iy) e^{iyt} dy \\ \sqrt{2\pi}f(-t) &= \int_{-\infty}^{\infty} F(iy) e^{-iyt} dy\end{aligned}$$

Or on replacing t by w ,

$$\sqrt{2\pi}f(-w) = \int_{-\infty}^{\infty} F(iy) e^{-iyw} dy$$

Given that

$$\begin{aligned}\mathcal{F}\{F(iy)\} &= \sqrt{2\pi}f(-w) \\ \mathcal{F}\{f(t)\} &= F(i\alpha)\end{aligned}$$